

Experiment No.2

Aim: To study and implement Hosted Virtualization using Virtual Box and KVM.

Theory:

A virtual machine functions as a software-based computer, mirroring the attributes of a physical machine by running an operating system and applications. Facilitated by a hypervisor, installed on the physical hardware within a virtualized data center, virtual machines possess their own virtualized hardware components, including CPU, memory, hard disk, and network interface card.

In a vSphere environment, ESXi serves as the hypervisor, dynamically allocating physical hardware resources to virtual machines as required. This abstraction grants virtual machines a level of autonomy from the underlying physical hardware, enabling seamless mobility across hosts and storage types. Virtualization, by decoupling virtual machines from specific hardware, allows for the consolidation of computing resources into flexible pools, managed effectively with tools like vCenter Server.

Overview:

1. **Resource Consolidation:** Virtualization aims to consolidate physical resources such as CPUs, memory, storage, and network components into pooled resources. This optimizes utilization and improves efficiency by allowing multiple virtual machines to share the same underlying hardware.
2. **Isolation and Independence:** Virtualization provides isolation between virtual machines, ensuring that the operations of one VM do not impact others. Each VM operates independently, with its own virtualized hardware components, enabling flexibility and reducing dependencies on specific physical hardware.
3. **Mobility and Flexibility:** Virtualization allows for the easy mobility of virtual machines. VMs can be migrated from one physical host to another or have their virtual disks moved between different storage types. This flexibility enhances adaptability to changing infrastructure requirements and simplifies maintenance tasks.
4. **Improved Resource Management:** Virtualization enables dynamic allocation and deallocation of resources based on the needs of virtual machines. This ensures efficient resource utilization, as the hypervisor adjusts resource allocations in real-time to meet varying workloads and demands.
5. **Enhanced Disaster Recovery and Security:** Virtualization enhances disaster recovery capabilities by enabling features such as snapshotting, cloning, and rapid VM deployment. Additionally, security is improved as virtual machines operate independently, reducing the risk of vulnerabilities spreading between systems. The decoupling of VMs from specific hardware also facilitates backup and recovery processes.

Virtual Machines

A virtual machine program functions as software that creates a self-contained virtual computer system with simulated hardware devices. Operating within a window on the host operating system, this virtual machine "machine" allows users to run operating system installers or live CDs, tricking them into thinking they're on a real computer. The virtual machine's operating system is stored on a virtual hard drive, eliminating the need for partitioning. Although virtual machines introduce some performance overhead, making them slightly slower than native installations on physical hardware, they provide a convenient solution for running different operating systems within a window on the current desktop.

While less suitable for resource intensive tasks like demanding games requiring high graphics and CPU power, virtual machines offer a practical approach for various scenarios, such as testing new operating systems or running applications on different platforms.

Creating and running virtual machines

Step 1: Download Ubuntu ISO

- Go to the Ubuntu Downloads page.
- Download the desktop version of Ubuntu.

Step 2: Install Oracle VirtualBox

- Go to the VirtualBox Downloads page.
- Download the appropriate version for your operating system and follow the installation instructions.

Step 3: Create a New Virtual Machine

- Open VirtualBox and click on "New."
- Name your virtual machine, select "Linux" as the type, and choose "Ubuntu" as the version.
- Assign memory (at least 2GB is recommended).

Step 4: Create a Virtual Hard Disk

- In the virtual machine settings, go to "Storage."
- Under "Controller: IDE," click on "Empty" under "Attributes."
- In the right panel, click on the disk icon next to "Optical Drive," then "Choose a disk file."
- Select the Ubuntu ISO file you downloaded.

Step 5: Start the Virtual Machine

- With the virtual machine selected, click "Start."
- The system will boot from the Ubuntu ISO, initiating the installation process.

Step 6: Install Ubuntu

- Follow the on-screen instructions to install Ubuntu.
- When prompted, select "Erase disk and install Ubuntu" for simplicity.

Step 7: Complete the Installation

- Follow the prompts to set your location, keyboard layout, and create a user account.
- Once the installation completes, remove the ISO from the virtual optical drive and restart the virtual machine.

Installing and Configuring KVM on Ubuntu:

Step 1: Open Terminal

- Press Ctrl + Alt + T to open a terminal window.

Step 2: Check Virtualization Support

- `egrep -c '(vmx|svm)' /proc/cpuinfo`

Ensure the output is greater than 0, indicating that your CPU supports virtualization.

Step 3: Install KVM

- `sudo apt update`
- `sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients virtinst.`

Step 4: Add User to libvirt Group

- `sudo adduser $USER libvirt`

Step 5: Verify KVM Installation

- `virsh list --all`

Verify that there are no errors and the virtualization environment is operational.

Step 6: Install Virt-Manager (Optional GUI)

- `sudo apt install virt-manager`
- Open Virt-Manager for a graphical interface to manage virtual machines.

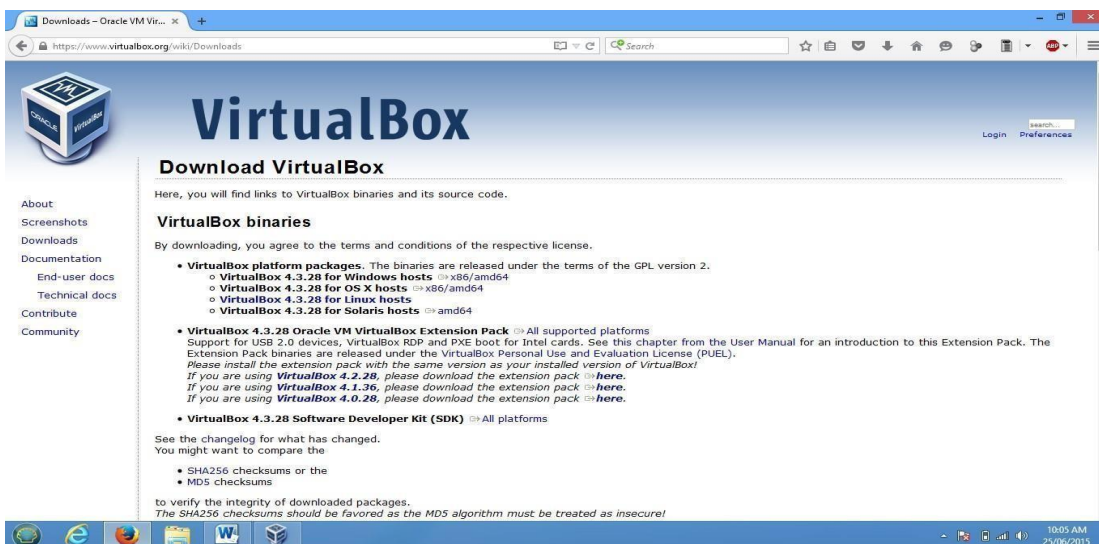
Step 7: Create a Virtual Machine with Virt-Manager (Optional)

- Open Virt-Manager and follow the GUI to create a new virtual machine.

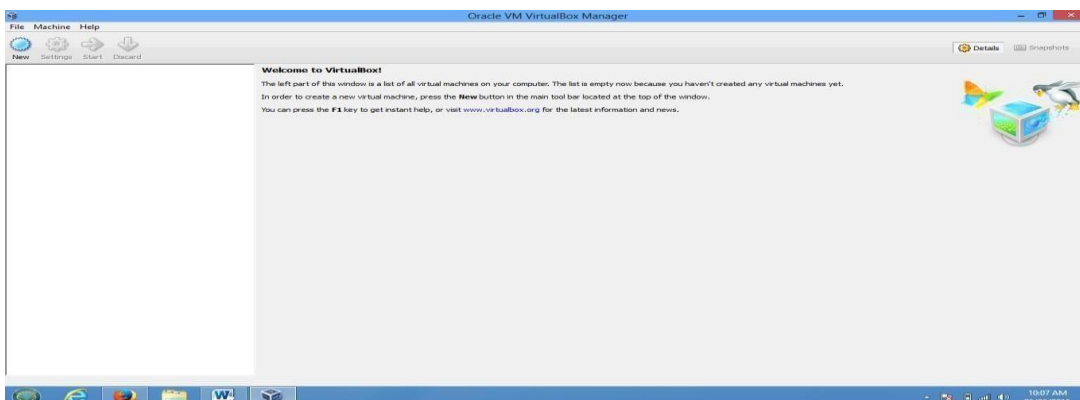
Step 8: Verify KVM Virtual Machine Operation

- Start the virtual machine and ensure it runs successfully.

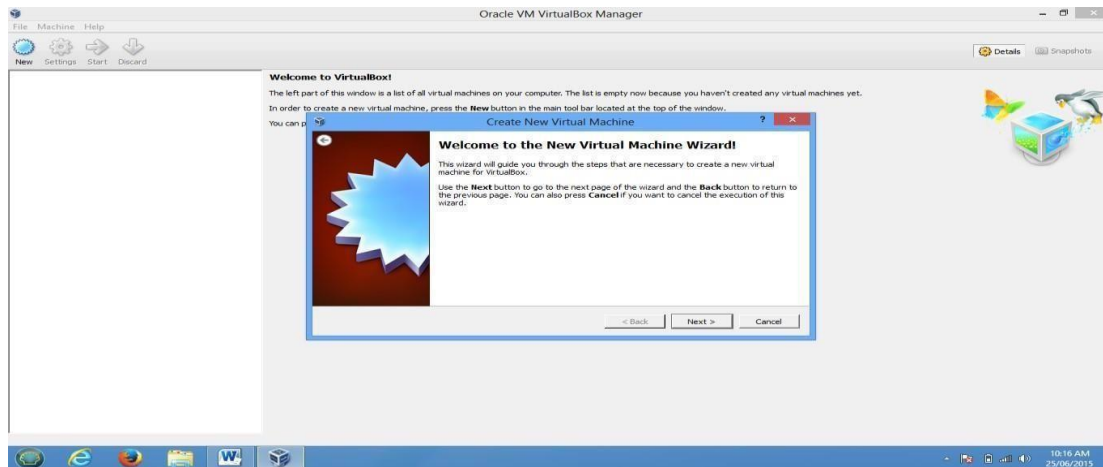
Step 1: Download Oracle Virtual box from <https://www.virtualbox.org/wiki/Downloads>



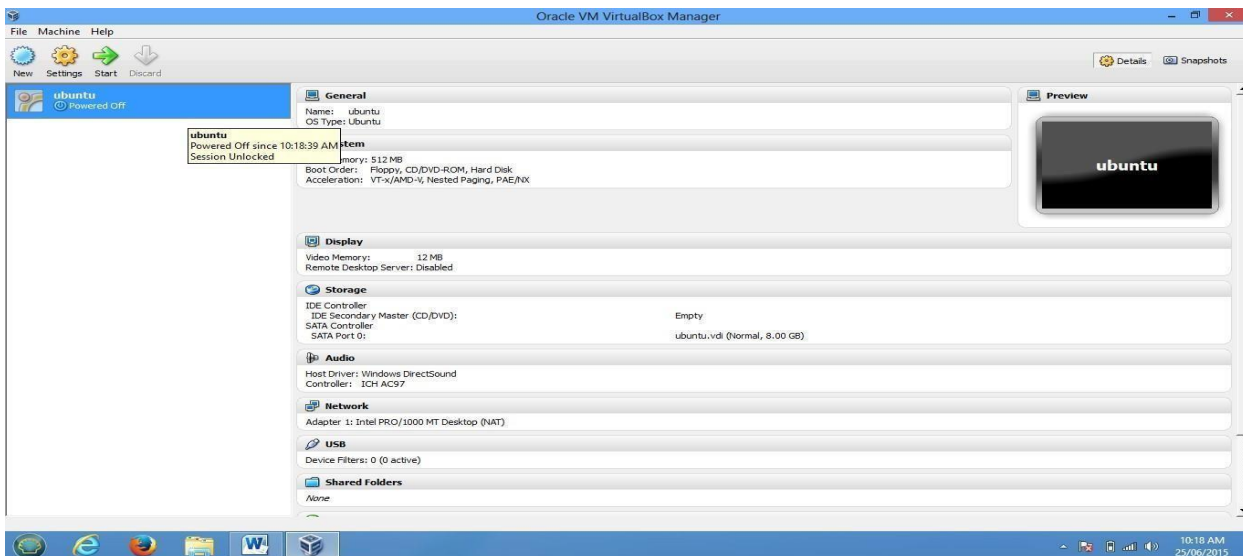
Step 2: Install it in Windows, Once the installation has done open it.



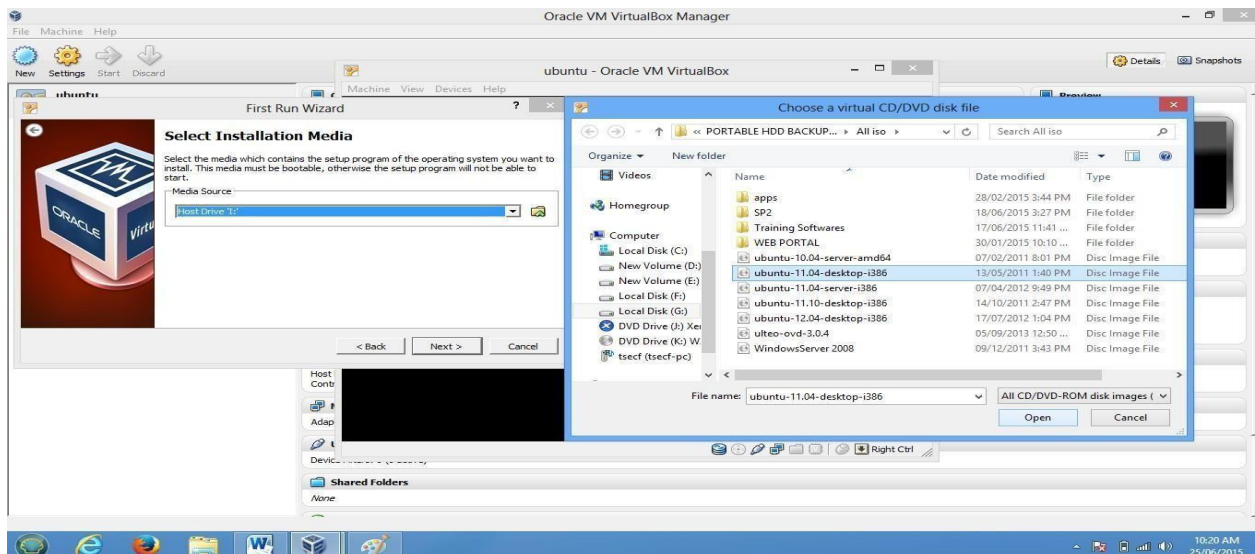
Step 3: Create Virtual Machine by clicking on New



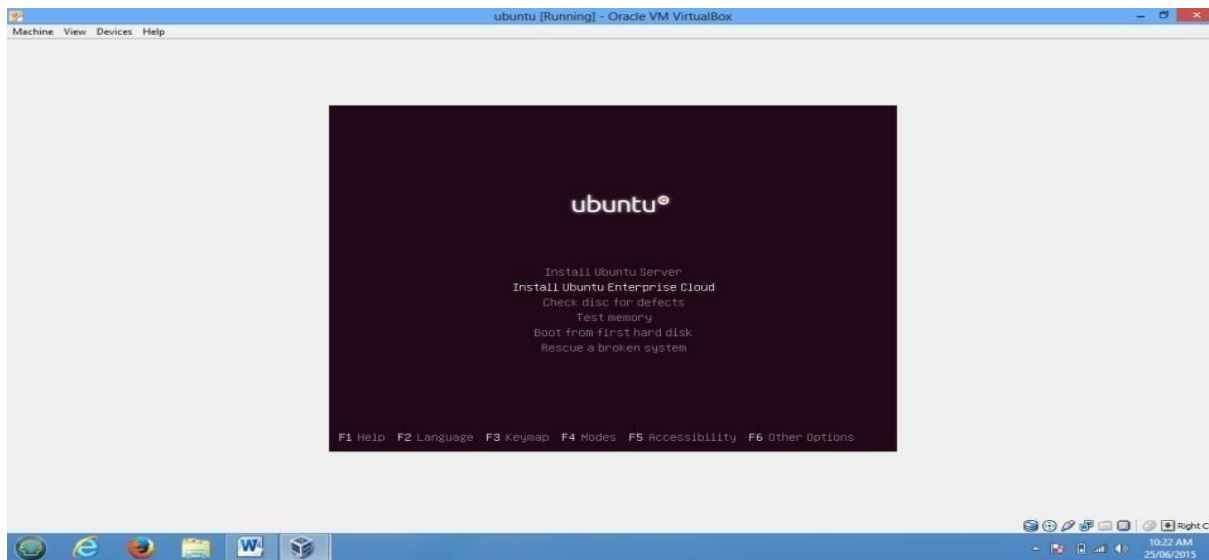
Step 4: Specify RAM Size, HDD Size, and Network Configuration and Finish the wizard



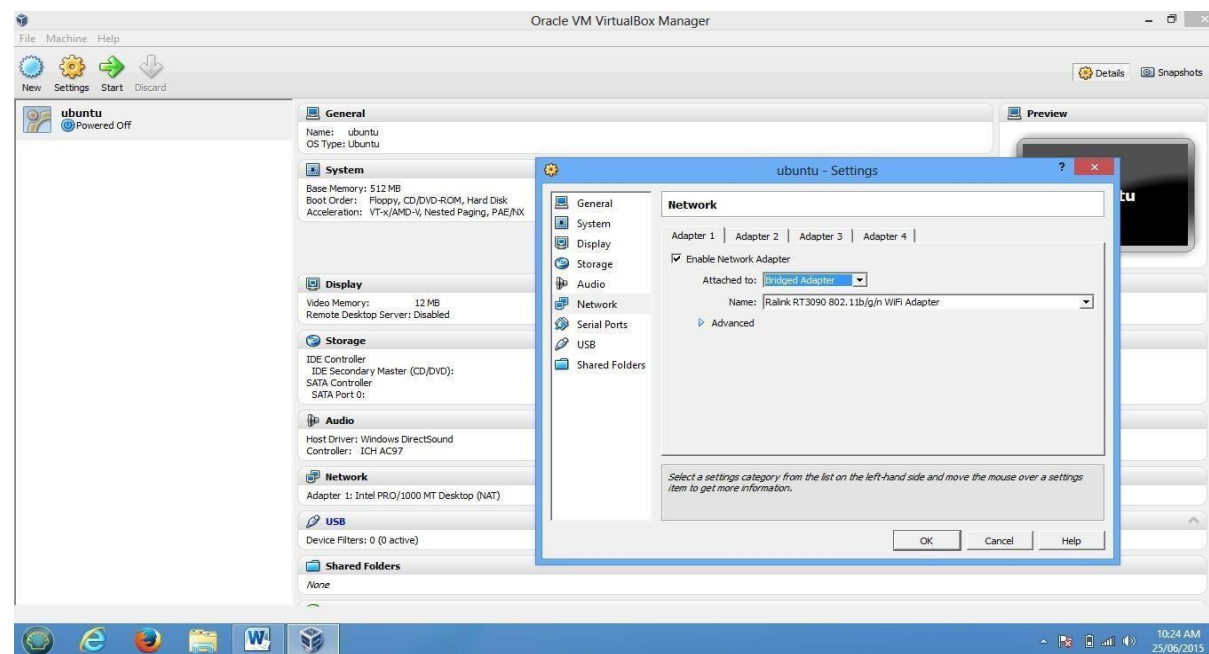
Step 5: To Select the media for installation, click on start and browse for ISO file



Step 6: Complete the Installation and use it.



Step 7: To Connect OS to the network change network mode to Bridge Adaptor



1) Hosted Virtualization on KVM Hypervisor

The Steps to Create and run Virtual machines in KVM are as follows

1. Check whether CPU has hardware virtualization support.

KVM only works if your CPU has hardware virtualization support – either Intel VT-x or AMDV. To determine whether your CPU includes these features, run the following command:

```
#sudo grep -c "svm\|vmx" /proc/cpuinfo
```

```
root@ubuntu:/home/tsec# sudo grep -c "svm\|vmx" /proc/cpuinfo
3
root@ubuntu:/home/tsec#
```

A 0 indicates that your CPU doesn't support hardware virtualization, while a 1 or more indicates that it does.

2. Install KVM and supporting packages.

Virt-Manager is a graphical application for managing your virtual machines. You can use the `kvm` command directly, but `libvirt` and Virt-Manager simplify the process.

```
#sudo apt-get install qemu-kvm libvirt-bin bridge-utils virt-manager
```

3. Create User

Only the root user and users in the `libvirtd` group have permission to use KVM virtual machines. Run the following command to add your user account to the `libvirtd` group:

```
#sudo adduser tsec #sudo adduser tsec libvirtd
```

After running this command, **log out** and **log back in as tsec**

```
root@ubuntu:/home/tsec# adduser tsec
root@ubuntu:/home/tsec# adduser tsec libvirtd
```

4. Check whether everything is working correctly.

Run following command after logging back in **as tsec** and you should see an empty list of virtual machines. This indicates that everything is working correctly.

```
#virsh -c qemu:///system list
```

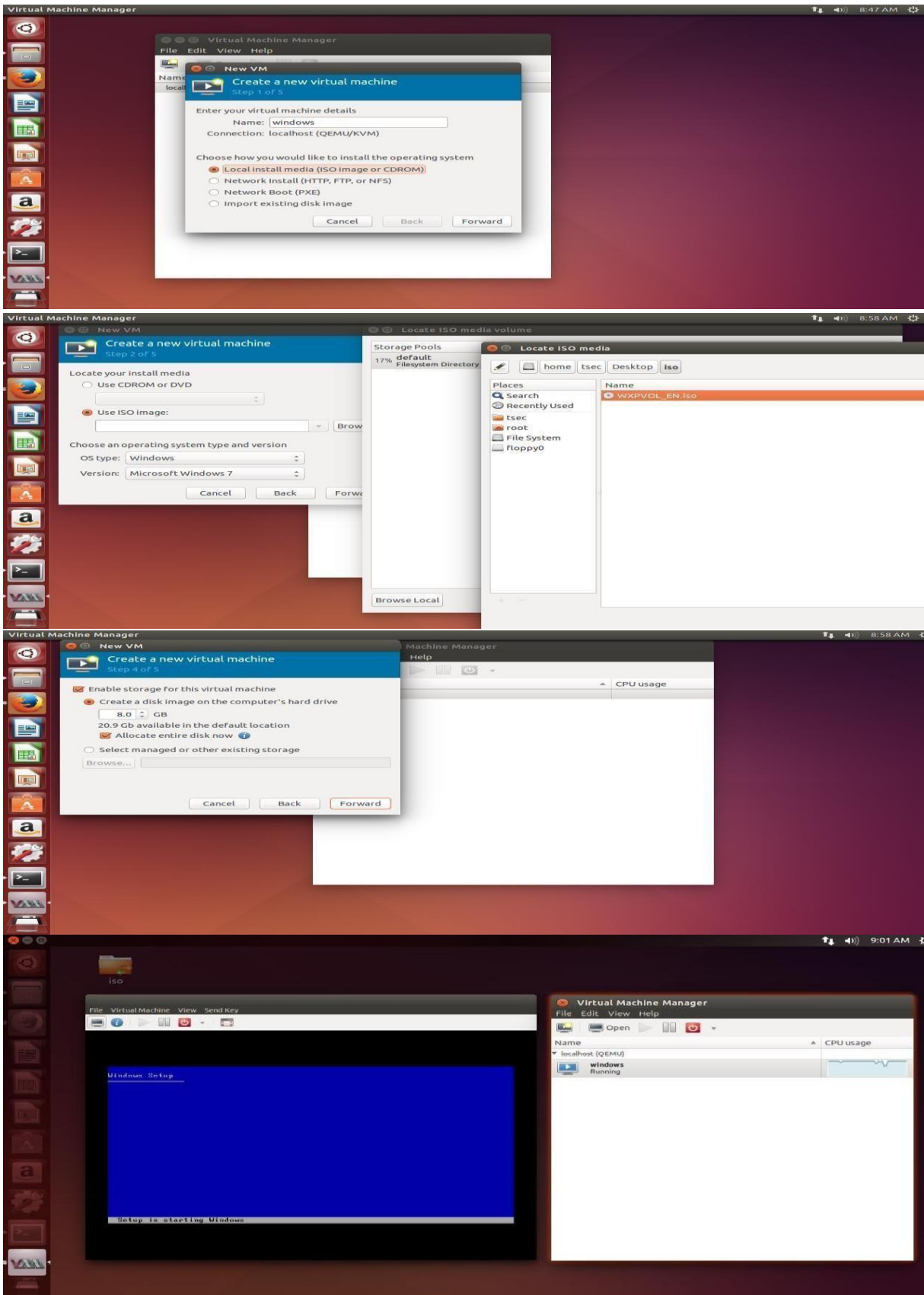
```
root@ubuntu:/home/tsec# virsh -c qemu:///system list
 Id      Name
-----
root@ubuntu:/home/tsec#
```

5. Open Virtual Machine Manager application and Create Virtual Machine

```
#virt-manager
```

```
root@ubuntu:/home/tsec# virt-manager
```

6. Create and run Virtual Machines



```
root$ sudo virsh list --all
```

Id	Name	State

1	ubuntu-vm	running

```
root$ virsh
```

```
Welcome to virsh, the virtualization interactive terminal.
```

```
Usage:  'help' for help with commands  
        'quit' to quit
```

```
virsh #
```

```
virsh #
```

```
virsh # list --all
```

Id	Name	State

1	Windows	running

Conclusion: Thus, we successfully demonstrated the installation of Ubuntu using Oracle VirtualBox and subsequently configured KVM on the Ubuntu system, providing a comprehensive hands-on experience in virtualization technologies